آغا خان یونیورسٹی ایگزامینیشن بورڈ

AGA KHAN UNIVERSITY EXAMINATION BOARD

Teaching and Learning Resources for Grade X Physics

Recommended Key Textbook:

Physics 10, Punjab Curriculum and Textbook Board (2012), Lahore

Additional Recommended Resource Material				
Topic No.	**Topic Title**	**Reference Book**	**Topic in the Reference Book**	**Website/ Link**
11	Waves Motion	John Avison, (2'd Ed) (2014). *The World of Physics*. Nelson.	Unit 20: Vibrations and Waves	http://phet.colorado.edu/ http://www.123physics.com http://www.physicsclassroom.com/class https://ebooks.stbb.edu.pk/english/class-x
		Duncan, Tom and Kennett, Heather. South Asia Edition, *Cambridge IGCSE Physics*, (3'd Ed) (2014). Hodder Edition.	Unit 2: Waves and Sound	
		Abbot, A. F. (1989). *Physics*. India: Heinemann Educational Publishers.	Unit 27: Longitudinal Waves and Sound	
		Charles Chew, Leong See Cheng, Chow Siew Foong. *Physics A Course of O Level*, Second Edition (2004).	Unit 13: General Wave Properties	
		Malik Sirajuddin and Sons. *Physics X*, Lahore, (2013)	Unit 10: Simple Harmonic Motion and Waves	

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		Sindh Textbook Board (2022). *The Textbook of Physics for Classes X*. Jamshoro: Sindh Textbook Board.	Unit 10: General Waves Properties	

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Topic No.	Topic Title	Reference Book	Topic in the Reference Book	Website/ Link
12	Sound	John Avison, (2'd Ed) (2014). *The World of Physics*. Nelson.	Unit 21: Sound Waves	http://phet.colorado.edu/ http://www.physicsclassroom.com/class https://ebooks.stbb.edu.pk/english/class-x
		Duncan, Tom and Kennett, Heather. South Asia Edition, *Cambridge IGCSE Physics*, (3'd Ed) (2014). Hodder Edition.	Unit 2: Waves and Sound	
		Abbot, A. F. (1989). *Physics*. India: Heinemann Educational Publishers.	Unit 27: Longitudinal Waves and Sound Unit 28: Musical Sounds	
		Charles Chew, Leong See Cheng, Chow Siew Foong. *Physics A Course of O Level*, Second Edition (2004).	Unit 16: Sound	
		Malik Sirajuddin and Sons. *Physics X*, Lahore, (2013)	Unit 11: Sound	
		Sindh Textbook Board (2022). *The Textbook of Physics for Classes X*. Jamshoro: Sindh Textbook Board.	Unit 11: Sound	
13	Geometrical Optics	John Avison, (2'd Ed) (2014). *The World of Physics*. Nelson.	Unit 2: Light Rays and Refraction Unit 3: Lenses and Optical Instruments	http://phet.colorado.edu/ http://www.123physics.com http://www.physicsclassroom.com/class https://ebooks.stbb.edu.pk/english/class-x
		Duncan, Tom and Kennett, Heather. South Asia Edition, *Cambridge IGCSE Physics*, (3'd Ed) (2014). Hodder Edition.	Unit 1: Light and Sight	

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		Abbot, A. F. (1989). *Physics*. India: Heinemann Educational Publishers.	Unit 21: Light Rays and Reflection of Light Unit 22: Spherical Mirrors Unit 23: Refraction at Plane Surfaces Unit 24: Lenses and Optical Instruments	
		Malik Sirajuddin and Sons. *Physics X*, Lahore, (2013)	Unit 12 Geometrical Optics	
		Sindh Textbook Board (2022). *The Textbook of Physics for Classes X*. Jamshoro: Sindh Textbook Board.	Unit 13: Geometrical Optics	

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14	Electrostatics	John Avison, (2'd Ed) (2014). *The World of Physics*. Nelson.	Unit 11: Static Electrons	http://phet.colorado.edu/ http://www.123physics.com http://www.physicsclassroom.com/class https://ebooks.stbb.edu.pk/english/class-x
		Duncan, Tom and Kennett, Heather. South Asia Edition, *Cambridge IGCSE Physics*, (3'd Ed) (2014). Hodder Edition.	Unit 7: Electricity and Electromagnetic Effects	
		Abbot, A. F. (1989). *Physics*. India: Heinemann Educational Publishers.	Unit 32: Static Electricity	
		Malik Sirajuddin and Sons. *Physics X*, Lahore, (2013)	Unit 16: Electrostatics	
		Sindh Textbook Board (2022). *The Textbook of Physics for Classes X*. Jamshoro: Sindh Textbook Board.	Unit 14: Electrostatics	
15	Electric Current	John Avison, (2'd Ed) (2014). *The World of Physics*. Nelson.	Unit 12: Electric Current	http://phet.colorado.edu/ http://www.123physics.com http://www.physicsclassroom.com/class https://ebooks.stbb.edu.pk/english/class-x
		Duncan, Tom and Kennett, Heather. South Asia Edition, *Cambridge IGCSE Physics*, (3'd Ed) (2014). Hodder Edition.	Unit 7: Electricity and Electromagnetic Effects	
		Abbot, A. F. (1989). *Physics*. India: Heinemann Educational Publishers.	Unit 35: Current, Electromotive Force and Resistance	
		Charles Chew, Leong See Cheng, Chow Siew Foong. *Physics A Course of O Level*, Second Edition (2004).	Unit 18: Current Electricity	

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		Malik Sirajuddin and Sons. *Physics X*, Lahore, (2013)	Unit 14: Current Electricity	
		Sindh Textbook Board (2022). *The Textbook of Physics for Classes X*. Jamshoro: Sindh Textbook Board.	Unit 15: Current Electricity	

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16	D.C. Circuits	Charles Chew, Leong See Cheng, Chow Siew Foong. *Physics A Course of O Level*, Second Edition (2004).	Unit 19: DC Circuit	http://phet.colorado.edu/ https://ebooks.stbb.edu.pk/english/class-x
		Malik Sirajuddin and Sons. *Physics X*, Lahore, (2013)	Unit 14: Current Electricity	
		Sindh Textbook Board (2022). *The Textbook of Physics for Classes X*. Jamshoro: Sindh Textbook Board.	Unit 15: Current Electricity	
17	Magnetism and Electromagnetism	John Avison, (2'd Ed) (2014). *The World of Physics*. Nelson.	Unit 14: Magnetism and Electromagnetism	http://phet.colorado.edu/ http://www.123physics.com http://www.physicsclassroom.com/class https://ebooks.stbb.edu.pk/english/class-x
		Duncan, Tom and Kennett, Heather. South Asia Edition, *Cambridge IGCSE Physics*, (3'd Ed) (2014). Hodder Edition.	Unit 7: Electricity and Electromagnetic Effects	
		Abbot, A. F. (1989). *Physics*. India: Heinemann Educational Publishers.	Unit 31: Magnetic Fields Unit 36: Magnetic Effect of an Electric Current Unit 38: The Principle of the Electric Motor	
		Charles Chew, Leong See Cheng, Chow Siew Foong. *Physics A Course of O Level*, Second Edition (2004).	Unit 21: Simple Phenomena of magnetism Unit 22: Force on Conductor in a Magnetic Field	
		Malik Sirajuddin and Sons. *Physics X*, Lahore, (2013)	Unit 15: Electromagnetism	

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		Sindh Textbook Board (2022). *The Textbook of Physics for Classes X*. Jamshoro: Sindh Textbook Board.	Unit 16: Electromagnetism	
		CANTAB Publisher, Lahore. *Model Textbook of Physics for Grade 9.* Based on National Curriculum of Pakistan 2022-23	Unit 11: Magnetism	
18	Electromagnetic Induction	John Avison, (2'd Ed) (2014). *The World of Physics*. Nelson.	Unit 14: Magnetism and Electromagnetism	http://phet.colorado.edu/
		Charles Chew, Leong See Cheng, Chow Siew Foong. *Physics A Course of O Level*, Second Edition (2004).	Unit 23: Electromagnetic Effects	
		Malik Sirajuddin and Sons. *Physics X*, Lahore, (2013)	Unit 15: Electromagnetism	
19	Electromagnetic Waves and Electronics	Duncan, Tom and Kennett, Heather. South Asia Edition, *Cambridge IGCSE Physics*, (3'd Ed) (2014). Hodder Edition.	Unit 02: Waves and Sound	https://scied.ucar.edu/learning-zone/earth-system/electromagnetic-spectrum https://steemit.com/steemstem/@shehzad/basics-of-production-transmission-and-reception-of-electromagnetic-waves-radio-waves https://ebooks.stbb.edu.pk/english/class-x
		Abbot, A. F. (1989). *Physics*. India: Heinemann Educational Publishers.	Unit 45: Electrons in Solids, and Semiconductor Devices	
		Charles Chew, Leong See Cheng, Chow Siew Foong. *Physics A Course of O Level*, Second Edition (2004).	Unit 24: Introductory Electronics	
		Malik Sirajuddin and Sons. *Physics X*, Lahore, (2013)	Unit 16: Electronics	

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		Sindh Textbook Board (2022). *The Textbook of Physics for Classes X*. Jamshoro: Sindh Textbook Board.	Unit 17: Introductory Electronics	

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20	Atomic and Nuclear Physics	John Avison, (2'd Ed) (2014). *The World of Physics*. Nelson.	Unit 18: Radioactivity	http://phet.colorado.edu/ https://ebooks.stbb.edu.pk/english/class-x
		Duncan, Tom and Kennett, Heather. South Asia Edition, *Cambridge IGCSE Physics*, (3'd Ed) (2014). Hodder Edition.	Unit 8: Electrons and Atoms	
		Abbot, A. F. (1989). *Physics*. India: Heinemann Educational Publishers.	Unit 46: Atomic Structure, the Quantum Theory and Radioactivity	
		Charles Chew, Leong See Cheng, Chow Siew Foong. *Physics A Course of O Level*, Second Edition (2004).	Unit 25: Radioactivity and the Nuclear Atom	
		Malik Sirajuddin and Sons. *Physics X*, Lahore, (2013)	Unit 19: Atomic Structure Unit 20: Nuclear Structure	
		Sindh Textbook Board (2022). *The Textbook of Physics for Classes X*. Jamshoro: Sindh Textbook Board.	Unit 9: Nuclear Physics	

Note: This resource list has been prepared primarily for teachers. While it can be shared with students, students should not be required to buy multiple books. Schools are encourages to stock these books in the library. Moreover, these are only suggestions which have been compiled for the ease of teachers and students; schools are encouraged to use other resources for teaching and learning as well, as long as they are in line with the student learning outcomes (SLOs) mentioned in AKU-EB syllabi. Unless specified, AKU-EB does not endorse any of these books or websites. You are advised to use an ad-blocker while accessing the websites. In case any website is not functional for any reason, you may inform us at examination.board@aku.edu for an alternative or search material via any search engine. If you have any query, please contact us via email.

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